CO3 AT1

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1714**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I SRINIVASAN.A 192411025 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A. Srinij

22+22+17+18=78

87

CO3 A11

Cybersecurity IDS

(a) Propositional Logic.

Let:

* F = Failed Login Attempts
* U = UnusualIPAdress
* O = oddLoginTime
* H = HighDataTransfer
* B = Brute force Attack
* C = Account compromise
* D = Data Exfiltration

Facts: F, U, H are true.

(b) Forward chaining

* $F \wedge U \rightarrow B \rightarrow$ Brute Force Attack
* $U \wedge H \rightarrow D \rightarrow$ Data Exfiltration
* $F \wedge O \rightarrow C$ cannot fire because O is not given.

Desired: B, D.

(c) Backward chaining

Goal: D

D requires $U \wedge H$ using Rule 3.

Both U and H are facts.

Therefore, D is proved.

D
$\downarrow$
$U \wedge H$
$\downarrow \quad \downarrow$

9. Airport priority Landing.

(a) Unification

Rule 1:

$A(x) \land E(x) \rightarrow P(x)$

Given:

A(flightZ), E(flightZ)

Substitution:

$\theta = \{x/flightZ\}$

Therefore:

P(flightZ)

(b) Resolution

Relevant clauses:

1. $\neg A(x) \lor \neg E(x) \lor P(x)$
2. $\neg P(x) \lor R(x)$
3. $\neg A(x) \lor \neg Q(x,y) \lor \neg R(y) \lor D(x)$

Facts:

A(Z), E(Z), A(Y), Q(Y, Z)

Hence Delay (flight Y) is true.

(c) Simultaneous emergencies

current KB can identify multiple emergency aircraft, but cannot properly handle runway conflicts.

Industrial Maintenance

Let:

* M = MachineOverheating
* N = MaintenanceNeeded
* H = HighPriorityLine
* S = ScheduleImmediateRepair
* R = ProductionRisk

(a) CNF

P1:

$M \rightarrow N$

$\rightarrow \neg M \vee N$

P2:

$N \wedge H \rightarrow S$

$\rightarrow \neg N \vee \neg H \vee S$

P3:

$\neg S \wedge H \rightarrow R$

$\rightarrow S \vee \neg H \vee R$

(b) Resolution

Goal: S

Negated goal: $\neg S$

1. $\neg M \vee N + M \rightarrow N$
2. $\neg N \vee \neg H \vee S + N \rightarrow \neg H \vee S$
3. $\neg H \vee S + H \rightarrow S$
4. $S + \neg S \rightarrow \square$

∴ ScheduleImmediateRepair is proved.

(c) Remove M:

without M:

* N cannot be derived
* S cannot be derived

* R also cannot be derived because $\neg S$ is not given.

4. Minefield Robot.

(a) Belief state

Given :

* vibration [2, 1]
* Smoke [1, 1]
* Beacon [2, 2]

(b) Forward chaining

vibration [2, 1]

↓

Mine adjacent [2,]

↓

Smoke [1, 1]

↓

Fire adjacent [1, 1]

↓

Beacon [2, 2]

↓

Safezone Marker [2, 2]

(c) Path safety

Path :

[1, 1] → [2, 1] → [2, 2]

* [1, 1] → Fire hazard nearby
* [2, 1] → Mine hazard nearby
* [2, 2] → Safe-Zone marker, but no necessarily proven safe.

Conclusion : The path cannot be confirmed as safe with the given KB.